

Battery Charger Flash MCU

Part number	Internal clock	VDD	System clock	Program memory	Data memory	Data EEPROM	I/O	Interface	Timer	A/D	D/A	OPA	Package
HT45F5Q-1	8MHz	2.2V~5.5V	8MHz	1K×14	32×8	32×14 [#]	9	—	—	10-bit×5	8-bit×1 12-bit×1	2	16NSOP
HT45F5Q-2	8MHz 32kHz	2.2V~5.5V	125kHz~8MHz or 32kHz	2K×16	128×8	32×8 [#]	15	UART×1	10-bit CTM×1	12-bit×7	8-bit×1 12-bit×1	3	20NSOP
HT45F5Q-3	8MHz 32kHz	2.2V~5.5V	125kHz~8MHz or 32kHz	4K×15	256×8	32×15 [#]	23	SPI/1 ² C/ UART×1	10-bit CTM×1 10-bit STM×1	12-bit×11	14-bit×1 12-bit×1	3	24/28SSOP

Fig. 2: HT45F5Q-X specifications

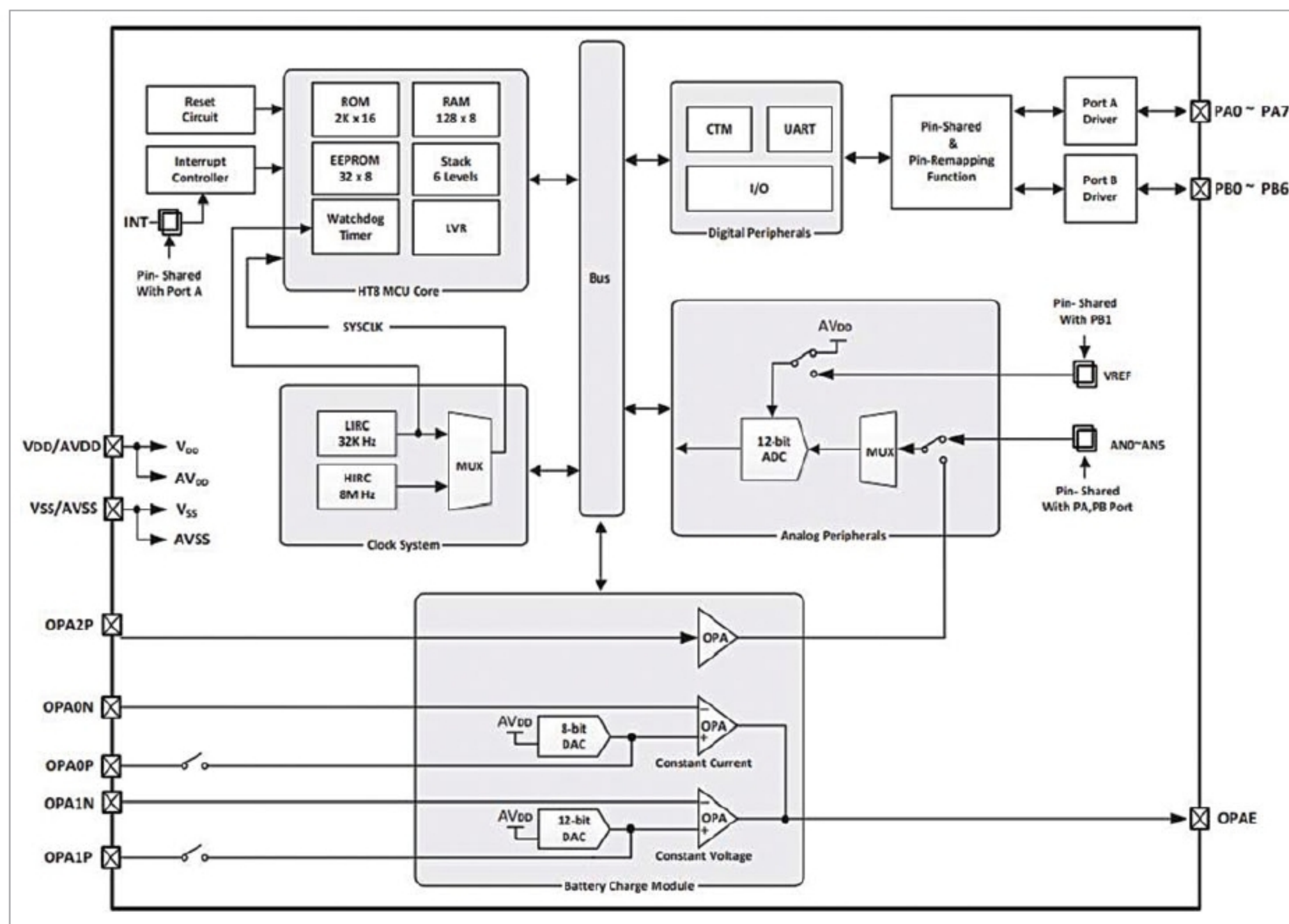


Fig. 3: Block diagram of HT45F5Q-2

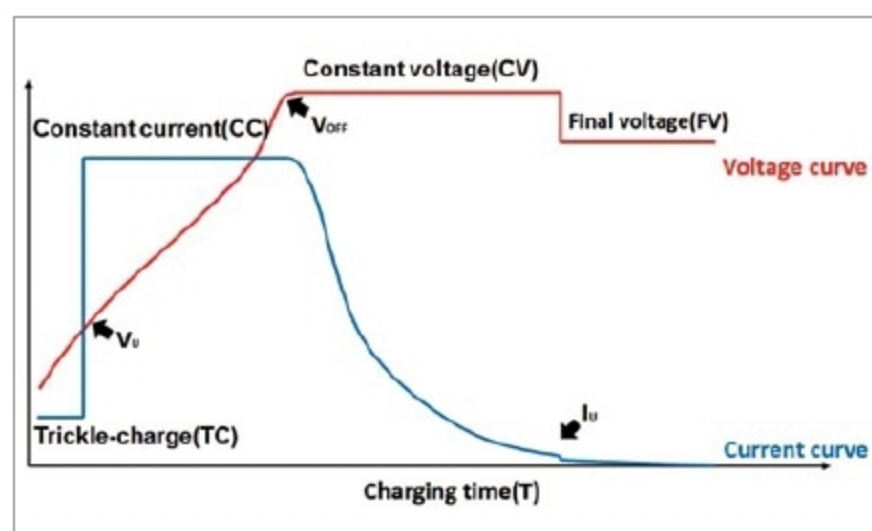


Fig. 4: Battery charging curve

module, which can be utilised for closed-loop charging control with constant voltage and constant current for efficiently charging a battery. Internal block diagram of MCU HT45F5Q-2 is shown in Fig. 3.

The battery charging module in HT45F5Q-2 has built-in OPAs and DACs

that are needed for charging process. Therefore the design reduces the need for external components like shunt regulators, OPAs and DACs, which are commonly used in conventional battery charging circuits. As a result, the peripheral circuit is compact and simple, resulting in a smaller PCB area and low overall cost.

Working of EV charger

Input power to the EV charger is an AC voltage in the range of 170V to 300V. The EV charger uses a half-bridge LLC